Exercice 2 :

a/ $\frac{1}{1+j(\frac{\omega}{R}L)-\frac{1}{R\omega C}} = \frac{H_0}{1+jQ(\frac{\omega}{\omega_0}-\frac{\omega_0}{\omega})}$

$H_0 = 1$

$\begin{cases} \frac{L}{R} = \frac{Q}{\omega_0}.1 \\ \frac{1}{RC} = \omega_0 Q.2 \end{cases} \quad \begin{cases} \frac{2}{1} \rightarrow \frac{1}{LC} = \omega_0^2 \\ \Rightarrow \omega_0 = \frac{1}{\sqrt{LC}} \\ Q = \frac{1}{R}\sqrt{\frac{L}{C}} \end{cases}$

$\omega_0 = 5000 rad.s^{-1}$

$Q = 100$

$\omega_0 = 3\omega$

b/

$$U_s(t) = -\frac{8A}{\pi^2}\sum_{k=0}^{+\infty}\frac{|H(j(2k+1)\omega)|}{(2k+1)^2}cos((2k+1)\omega t + arg(\underline{H}(j(2k+1))\omega))$$

c/

$$A_k = -\frac{8A}{\pi^2(2k+1)^2} \times \frac{1}{\sqrt{1+Q^2((2k+1)\frac{\omega}{\omega_0}-\frac{\omega_0}{(2k+1)\omega})}}$$

$$\varphi_k = -arctan(Q(\frac{(2k+1)\omega}{\omega_0}-\frac{\omega_0}{(2k+1)\omega}))$$

d/

$$A_0 \simeq 0,03V$$
$$A_1 \simeq 0.90V$$
$$A_2 \simeq 0.0V$$
$$\varphi_0 \simeq -1.58rad$$
$$\varphi_1 \simeq rad$$
$$\varphi_2 \simeq 1.58rad$$

e/ $u_s \simeq 0.90cos(3\omega t)$

Exercice 3

La réponse à un signal continu est un signal continu.
Pas de discontinuité lors de la discontinuité de l'entrée
Donc passe-bas.

$$H(\omega) = \frac{H_0}{1+\frac{j}{Q}\frac{\omega}{\omega_0}-\frac{\omega_0^2}{\omega^2}}$$

$$H_0 = 2$$
$$Q \simeq 5$$
$$u_p = \omega\sqrt{1-\frac{1}{2Q^2}}$$
$$\simeq \omega_0$$

$$\underline{s}(1+\frac{j\omega}{Q\omega_0}+\frac{(j\omega)^2}{\omega_0^2}) = H_0\underline{e}$$
$$s(t)+\frac{1}{Q\omega_0}\frac{ds}{dt}+\frac{1}{\omega_0^2}\frac{d^2s}{dt^2} = H_0 e(t)(=1, t>0)$$
$$\frac{d^2s}{dt^2}+\frac{\omega_0}{Q}\frac{ds}{dt}+\omega_0^2 s(t) = \omega_0^2 H_0$$
$$\Delta = \frac{\omega_0^2}{Q^2}-4\omega_0^2 < 0$$
$$r_{1,2} = -\frac{\omega_0}{2Q} \pm j\omega_0\sqrt{1-\frac{1}{4Q^2}}$$
$$s(t) = ae^{r_1 t}+be^{r_2 t}$$
$$s(t) = e^{-\frac{\omega_0}{2Q}t}(ae^{j\omega_p t}+be^{-j\omega_p t})+H_0-Acos(\omega_p t+\varphi)$$

$$s'(t) = H_0 + Ae^{-\frac{\omega_0}{2Q}t}cos(\omega_p t+\varphi)$$

t_1 : premier max de s(t)
t_2 : 2e –

$$s(t_1) = 2+Ae^{-\frac{\omega_0}{2Q}t_1} = 3.5$$
$$s(t_2) = 2+Ae^{-\frac{\omega_0}{2Q}(t_1+T_0)} = 2.8$$

$\begin{cases} Ae^{-\frac{\omega_0}{2Q}t_1} = 1.5 \\ Ae^{-\frac{\omega_0}{2Q}(t_1+T_0)} = 0.8 \end{cases}$

$\frac{1.5}{0.8} = e^{\frac{\omega_0 T_0}{2Q}} = e^{\frac{\pi}{Q}}$

$\frac{\pi}{Q} = ln(\frac{1.5}{0.8})$

$Q = 5$

Exercice 4

a/

$$\begin{aligned} e &= u_R + u_C \\ &= R(i_C + i_R) + u_C \\ &= RC\frac{du_C}{dt} + u_c + u_c \\ &= \frac{du}{dt} + \frac{2}{RC}u = \pm\frac{E}{RC} \end{aligned}$$

c/

$$\begin{aligned} j\omega\underline{U} + \frac{2\underline{u}}{RC} &= \frac{\underline{e}}{RC} \\ \frac{\underline{u}}{\underline{e}} &= \frac{1}{RC(j\omega + \frac{2}{RC})} \\ \underline{H}(\omega) &= \frac{1}{jRC\omega + 2} \\ \underline{H} &= \frac{\frac{1}{2}}{1 + j\frac{RC\omega}{2}} \end{aligned}$$

$$= \frac{H_0}{1 + j\frac{\omega}{\omega_c}} \begin{cases} H_0 = \frac{1}{2} \\ \omega_c = \frac{2}{RC} = 2\omega_0 \end{cases}$$

- $\omega <<< \omega_c \begin{cases} G_{dB} = 20log(\frac{1}{2}) \\ = -6 \\ \varphi = 0 \end{cases}$
- $\omega >> \omega_c$

$$\begin{aligned} G_{dB} &= 20log(\frac{H_0\omega_c}{\omega}) \\ &= -6 + 20log(\frac{\omega_c}{\omega}) \\ \varphi &= -\frac{\pi}{2} \end{aligned}$$

Exercice 5

1/

$$\begin{aligned} A_0 &= \alpha E \\ E_{eff} &= \sqrt{\alpha}E \end{aligned}$$

2/

$$\begin{aligned} A_1 &= \frac{|sin\alpha\pi|}{\pi}2E \\ &= 0.0100.2E \\ A_{17} &= \frac{|sin17\alpha\pi|}{17\pi}2E \\ &= 0.00953 \times 2E \\ \frac{A_1 - A_{17}}{A_1} &= \frac{0.00047}{0.01000} = 0.047 \end{aligned}$$

3/ On observe le gain du filtre en fonction de f

Exercice 6

$$\underline{H}(j\omega) = \frac{H_0}{1 + jQ(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega})}$$

$H_0 = 1$
$f_0 = 1000Hz$
Pour $\omega >> \omega_0$

$$\begin{aligned} \underline{H}(j\omega) &\approx \frac{H_0}{jQ\frac{\omega}{\omega_0}} \\ G &= \frac{1}{Q\frac{\omega}{\omega_0}} \\ G_{dB} &= -20log(Q\frac{\omega}{\omega_0}) \\ -20 &= -20logQ \\ Q &= 10 \\ Q &= \frac{1}{R}\sqrt{\frac{L}{C}} \end{aligned}$$